Technology Stack

Date	02 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice. Technology Stack Details

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	*HTML5, CSS3, Bootstrap*	Provides a responsive and user-friendly interface for entering applicant details and displaying prediction results.

2	**Backend / Server-Side**	Python, Flask	Flask connects the frontend with the trained machine learning model and processes prediction requests efficiently.
3	**Database / Data Storage**	CSV Dataset (application_record.csv & credit_record.csv)	Stores applicant information and credit history used for preprocessing, model training, and prediction.
4	**Cloud / Hosting / Deployment**	IBM Watson Machine Learning, Flask Local Server	**Flask Local Server** IBM Watson enables cloud deployment of the trained model, while Flask supports local deployment and testing.

5	**Version Control & CI/CD**	*Git, GitHub*	Manages source code, version history, and collaboration during project development.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib/Pickle, Anaconda, PyCharm	Used for data preprocessing, visualization, model training, evaluation, model serialization, and application development.